Identification, Inference, and R Bootstrap Options for a Scalar Triangular System

A consolidated self-contained guide for point-identified and set-identified cases

June 2026

Abstract

This consolidated report combines the earlier identification guide for a scalar triangular system with a practical inference and implementation guide for standard errors, confidence intervals, and bootstrap methods in R. The maintained empirical object is a linear triangular model with one outcome, one endogenous regressor, one scalar external or constructed instrument when such a variable is used, and a common set of controls. The first part translates the identification ideas in Lewbel (2012), Lee–Lewbel–Schennach–Zhang's `trigmm` paper, and Lewis's survey on higher-moment identification into this scalar setting. The second part focuses on the inferential problem: how to bootstrap the point-identified generated-instrument estimator, how to bootstrap scalar set-identified endpoints, how to reproduce a `trigmmset`-style second moment confidence construction in R, and which R packages provide reusable building blocks. The document is written to be usable without prior familiarity with partial identification, while retaining the formulas needed to implement the methods directly.

Contents

1 Executive summary

The target coefficient is the scalar structural effect of the endogenous regressor, denoted by θ. The most useful summary is the following.

Option	Main identifying information	What is identified in the scalar triangular model	Main practical cautions
Lewbel exact heteroskedasticity	A scalar Z satisfies $\operatorname{Cov}(Z, \varepsilon_1\varepsilon_2) = 0$ and $\operatorname{Cov}(Z, \varepsilon_2^2) \neq 0$.	Point identification: $\theta = \operatorname{Cov}(Z, W_1W_2)/\operatorname{Cov}(Z, W_2^2)$	Needs heteroskedastic relevance of the first-stage residual. With one instrument it is just identified, so overidentification tests are unavailable.

Option	Main identifying information	What is identified in the scalar triangular model	Main practical cautions
Lewbel bounded-invalidity set	Same scalar Z, but $\mathrm{Corr}(Z, \varepsilon_1\varepsilon_2)$ is allowed to be nonzero, bounded relative to $\mathrm{Corr}(Z, \varepsilon_2^2)$.	A finite interval for θ when the relevance covariance is nonzero and the slack is below one.	The bound depends on a user-chosen slack τ; sampling uncertainty requires an outer confidence construction or bootstrap.
Lee–Lewbel–Schennach–Zhang point identification (`trigmm`)	Common-factor errors $W_2 = u + v$, $W_1 = \alpha u + \theta v + r$ with independent components and enough non-Gaussianity.	Point identification of (α, θ) from two higher-cumulant equations $M_p(\alpha, \theta) = 0$; then all other coefficients follow from the common residualization map.	Requires the sign of the common-factor loading and nondegenerate higher cumulants. Nonlinear GMM can have local minima and weak higher-moment identification.
Lee–Lewbel–Schennach–Zhang sharp set	Common-factor structure without the non-Gaussianity point-identifying condition; uses the implied decomposition of observables into Gaussian and non-Gaussian independent parts.	A sharp population joint set for the latent common-factor parameters.	Conceptually important but hard to estimate and interpret; the Stata paper therefore reports a simpler covariance-based outer bound.
Lee–Lewbel–Schennach–Zhang set identification (`trigmmset`)	Same common-factor structure and sign of common-factor loading, but no non-Gaussianity; only second moments.	A joint set in (α, θ). For positive common-factor loading, $\theta \leq B_0 \leq \alpha$ and $(B_0 - \theta)(\alpha - B_0) \leq D_0$. The projection onto θ alone is generally one-sided, not finite.	The reported confidence set is conservative and is a joint region; a finite interval for θ alone needs an extra magnitude restriction on the common-factor loading.
Lee–Lewbel–Schennach–Zhang plus an external instrument	Add a conventional IV moment such as $\mathbb{E}[Z(W_1 - \theta W_2)] = 0$ to the internal non-Gaussian GMM moments.	Usually point identification, often overidentification, with a direct comparison between the external-IV and internal-moment estimates.	If the two information sources conflict, the overidentifying restrictions can reject but cannot by themselves say which assumption failed.

Option	Main identifying information	What is identified in the scalar triangular model	Main practical cautions
Variance-regime / Rigobon-style identification	A scalar state or instrument Z partitions the sample into variance regimes; the structural impact matrix is fixed while shock variances change nonproportionally.	In a square two-shock representation, the impact ratio for the labeled W_2 shock can be identified from regime covariance matrices; in the one-changing-shock case it is a simple covariance-difference ratio.	This is an alternative square-shock model, not the three-shock common-factor model. Needs a credible regime/state and shock labeling.
Parametric volatility and unconditional volatility moments	GARCH, Markov-switching, smooth transition, stochastic-volatility, or autocovariances of squared residuals provide additional covariance equations.	Identifies a square impact matrix and hence an impact ratio after normalization and labeling.	Strong functional-form or time-series assumptions; high-moment estimates can be noisy.
Non-Gaussian ICA / likelihood / higher-moment GMM	Independent or partly independent non-Gaussian shocks in a square decomposition of (W_2, W_1).	Identifies the square mixing matrix up to sign, scale, and column order; θ is an impact ratio for the correctly labeled column.	Needs shock labeling, sign normalization, and enough departure from Gaussianity. Standard Wald intervals are unreliable under weak non-Gaussianity.

The bottom line is that there are two direct methods for the exact Lewbel triangular regression problem. Lewbel's method uses the scalar instrument Z to construct a heteroskedasticity-based instrument. Lee–Lewbel–Schennach–Zhang instead do not need Z for their internal point or set identification; they use a common-factor decomposition of the errors. The methods surveyed by Lewis are valuable alternatives and diagnostics, but they become direct estimators of θ only after one changes the latent structure to a square two-shock representation and labels one shock as the shock whose effect on W_1 per unit of W_2 is the desired coefficient.

2 Target scalar triangular system and notation

The notation follows the supplied TeX note, specialized to one endogenous regressor. Let $Y_{1,t+1}$ be the outcome and let $Y_{2,t+1}$ be a scalar endogenous regressor. Let X_t be the common vector of controls used in both the outcome equation and the reduced form for the endogenous regressor. The triangular model is

$$Y_{1,t+1} = X_t^\top \beta_1 + \theta Y_{2,t+1} + \varepsilon_{1,t+1}, \tag{1}$$
$$Y_{2,t+1} = X_t^\top \beta_2^R + \varepsilon_{2,t+1}. \tag{2}$$

The coefficient of interest is the scalar θ. The errors may be correlated, so ordinary least squares in (1) is not generally causal.

Project both observed variables on the same X_t and define the residuals

$$W_{1,t+1} := Y_{1,t+1} - X_t^\top \beta_1^R, \qquad \beta_1^R := \left(\mathbb{E}[X_t X_t^\top]\right)^{-1} \mathbb{E}[X_t Y_{1,t+1}], \tag{3}$$
$$W_{2,t+1} := Y_{2,t+1} - X_t^\top \beta_2^R, \qquad \beta_2^R := \left(\mathbb{E}[X_t X_t^\top]\right)^{-1} \mathbb{E}[X_t Y_{2,t+1}]. \tag{4}$$

Under $\mathbb{E}[X_t \varepsilon_{1,t+1}] = \mathbb{E}[X_t \varepsilon_{2,t+1}] = 0$, the residualized system is

$$W_{1,t+1} = \theta W_{2,t+1} + \varepsilon_{1,t+1}, \qquad W_{2,t+1} = \varepsilon_{2,t+1}. \tag{5}$$

For any candidate value w of θ, the implied structural residual is

$$\varepsilon_{1,t+1}(w) := W_{1,t+1} - w W_{2,t+1}. \tag{6}$$

Once θ is known or restricted to a set Θ, the non-endogenous coefficients in the outcome equation are recovered by the same linear map used in the supplied note:

$$\beta_1(w) = \beta_1^R - \beta_2^R w. \tag{7}$$

Thus each candidate w for θ implies one and only one candidate coefficient vector $\beta_1(w)$. This is why a finite interval for θ is the central object in the scalar problem.

Throughout, Z_{t+1} denotes a single observed scalar that may be used as an external instrument, a heteroskedasticity-generating variable, a variance-regime indicator, or a shock-labeling variable depending on the identification strategy. The role of Z_{t+1} is strategy-specific; it is not automatically a valid instrument simply because it is observed.

3 Baseline: Lewbel's scalar heteroskedasticity method

Although the two new attached papers are the main focus, the Lewbel baseline is useful because it fixes the notation and clarifies what it means to have one endogenous regressor and one instrument.

3.1 Exact validity gives a point estimate

Lewbel's exact triangular condition is

$$\mathrm{Cov}(Z_{t+1}, \varepsilon_{1,t+1}\varepsilon_{2,t+1}) = 0, \qquad \mathrm{Cov}(Z_{t+1}, \varepsilon_{2,t+1}^2) \neq 0. \tag{8}$$

The first condition says that Z_{t+1} is not correlated with the product of the two structural errors. The second condition says that Z_{t+1} predicts the variance of the first-stage error. Therefore

$$H_{t+1} := (Z_{t+1} - \mathbb{E}Z_{t+1}) W_{2,t+1} \tag{9}$$

is a valid constructed instrument for $W_{2,t+1}$ in the residualized equation. Indeed,

$$\mathbb{E}[H_{t+1}\varepsilon_{1,t+1}] = \mathrm{Cov}(Z_{t+1}, \varepsilon_{1,t+1}\varepsilon_{2,t+1}) = 0, \tag{10}$$
$$\mathbb{E}[H_{t+1} W_{2,t+1}] = \mathrm{Cov}(Z_{t+1}, \varepsilon_{2,t+1}^2) \neq 0. \tag{11}$$

In the scalar residualized system this yields the closed-form population coefficient

$$\theta = \frac{\mathrm{Cov}(Z_{t+1}, W_{1,t+1}W_{2,t+1})}{\mathrm{Cov}(Z_{t+1}, W_{2,t+1}^2)}. \tag{12}$$

With controls, the same calculation is applied after residualizing both $Y_{1,t+1}$ and $Y_{2,t+1}$ on X_t, and the remaining coefficients follow from (7).

3.2 Bounded invalidity gives a scalar interval

The supplied TeX note generalizes the exact condition by allowing a bounded amount of contamination. In the scalar case, assume that for a known $\tau \in [0, 1)$,

$$|\mathrm{Corr}(Z_{t+1}, \varepsilon_{1,t+1}\varepsilon_{2,t+1})| \le \tau \left|\mathrm{Corr}(Z_{t+1}, \varepsilon_{2,t+1}^2)\right|. \tag{13}$$

Define the observable population quantities

$$L := \mathrm{Cov}(Z_{t+1}, W_{1,t+1}W_{2,t+1}), \qquad Q := \mathrm{Cov}(Z_{t+1}, W_{2,t+1}^2), \tag{14}$$

$$S_0 := \mathrm{Var}(W_{1,t+1}W_{2,t+1}), \qquad S_1 := \mathrm{Cov}(W_{2,t+1}^2, W_{1,t+1}W_{2,t+1}), \qquad S_2 := \mathrm{Var}(W_{2,t+1}^2). \tag{15}$$

For a candidate w, the relative-correlation bound becomes the quadratic inequality

$$(L - Qw)^2 - d\{S_0 - 2S_1 w + S_2 w^2\} \le 0, \qquad d := \tau^2 \frac{Q^2}{S_2}. \tag{16}$$

If $Q \neq 0$ and $\tau < 1$, the leading coefficient is $Q^2 - dS_2 = Q^2(1 - \tau^2) > 0$, so the set is a closed finite interval whenever the discriminant is nonnegative. Writing

$$a := Q^2 - dS_2, qquadb := -2LQ + 2dS_1, \qquad c := L^2 - dS_0, \tag{17}$$

the interval is

$$\Theta_{\mathrm{Lewbel}}(\tau) = \left[\frac{-b - \sqrt{b^2 - 4ac}}{2a}, \frac{-b + \sqrt{b^2 - 4ac}}{2a}\right]. \tag{18}$$

At $\tau = 0$, the interval collapses to the point in (12). As τ grows, the interval widens. The interpretation is simple: the more invalidity one is willing to tolerate in the constructed instrument, the less precisely θ is identified.

3.3 Sampling uncertainty

The exact Lewbel point estimator can be computed by two-stage least squares using X_t and $(Z_{t+1} - \bar{Z})\widehat{W}_{2,t+1}$ as instruments. Heteroskedasticity- robust or cluster-robust covariance matrices are standard. For the interval (18), a conservative confidence set can be obtained by replacing the population moments with sample moments and using a bootstrap or a moment-inequality outer construction. The attached Lee et al. paper provides a more explicit closed-form confidence construction for its own second-moment set, which is described in Section 5.

4 Direct option 1 from Lee–Lewbel–Schennach–Zhang: point identification by non-Gaussianity

The `trigmm` paper studies the same triangular shape as (1)–(2), but it does not use a Lewbel-style scalar Z to construct an instrument. Instead, it imposes a latent common-factor structure on the residuals. In the notation of this report,

$$W_{2,t+1} = u_{t+1} + v_{t+1}, \qquad W_{1,t+1} = \theta W_{2,t+1} + \beta_u u_{t+1} + r_{t+1}. \tag{19}$$

Equivalently,

$$W_{1,t+1} = \alpha u_{t+1} + \theta v_{t+1} + r_{t+1}, \qquad \alpha := \theta + \beta_u. \tag{20}$$

Here u is the unobserved common cause that creates endogeneity, v is the idiosyncratic first-stage shock, and r is the idiosyncratic outcome shock. The key assumptions are:

- u_{t+1}, v_{t+1}, and r_{t+1} are mutually independent, and in the implemented command they are also independent of X_t;
- the sign of $\beta_u = \alpha - \theta$ is specified by the researcher;
- sufficiently high moments exist and the relevant higher cumulants are nondegenerate.

The sign condition matters because replacing u by $-u$ changes the sign of the common-factor loading without changing the observed distribution. Economic theory must therefore decide whether the common factor moves Y_2 and Y_1 in the same direction or in opposite directions.

4.1 Cumulant notation

Let Φ_{W_2,W_1} be the log characteristic function of $(W_{2,t+1}, W_{1,t+1})$, and let Φ_{W_2} be the log characteristic function of $W_{2,t+1}$. Define joint cumulants by

$$\Phi^{k,l}_{W_2,W_1} := \left.\frac{\partial^{k+l}\Phi_{W_2,W_1}(\zeta,\xi)}{i^{k+l}\partial\zeta^k\partial\xi^l}\right|_{\zeta=0,\xi=0}, \qquad \Phi^{k}_{W_2} := \left.\frac{\partial^{k}\Phi_{W_2}(\zeta)}{i^{k}\partial\zeta^k}\right|_{\zeta=0}. \tag{21}$$

For ordinary intuition, $\Phi^2_{W_2}$ is the variance of W_2, $\Phi^3_{W_2}$ is its third cumulant, and higher orders measure higher-order departures from Gaussianity. Gaussian variables have all cumulants above order two equal to zero.

Because cumulants of independent sums add, the latent model implies, for any nonnegative integer p,

$$\Phi^{3+p}_{W_2} = \kappa^{3+p}_u + \kappa^{3+p}_v, \tag{22}$$

$$\Phi^{2+p,1}_{W_2,W_1} = \alpha\kappa^{3+p}_u + \theta\kappa^{3+p}_v, \tag{23}$$

$$\Phi^{1+p,2}_{W_2,W_1} = \alpha^2\kappa^{3+p}_u + \theta^2\kappa^{3+p}_v, \tag{24}$$

where κ^m_u and κ^m_v are the order-m cumulants of u and v. Eliminating these nuisance cumulants gives the observable moment equation

$$M_p(\alpha,\theta) := \Phi^{1+p,2}_{W_2,W_1} - \alpha^2\Phi^{3+p}_{W_2} - (\theta+\alpha)\left(\Phi^{2+p,1}_{W_2,W_1} - \alpha\Phi^{3+p}_{W_2}\right) = 0. \tag{25}$$

This is the scalar version of the moment condition implemented by `trigmm`.

4.2 Identification from two cumulant equations

Two different values of p give two equations in the two unknowns (α, θ). For example,

$$M_0(\alpha, \theta) = 0, \qquad M_1(\alpha, \theta) = 0 \tag{26}$$

can point identify (α, θ). The default `trigmm` specification uses $p = 0, 1$. The command can also use $p = 0, 1, 2$, which gives three moment equations for two parameters and therefore overidentifying restrictions.

For a pair $q < \widetilde{q}$, the essential nondegeneracy condition is that the higher cumulants are not lined up in a way that makes the two equations carry the same information. In the notation here, a convenient statement is

$$\Phi^{3+\widetilde{q}}_{W_2}\,\Phi^{2+q,1}_{W_2,W_1} \neq \Phi^{3+q}_{W_2}\,\Phi^{2+\widetilde{q},1}_{W_2,W_1}. \tag{27}$$

For the default pair $q = 0$, $\widetilde{q} = 1$, this condition can fail if either u or v is Gaussian, if both are symmetric, or if they have exactly the same distribution. This is why the method is described as identification by non-Gaussianity: information must be present beyond means and variances.

Once (α, θ) is identified, the loading of the common factor is

$$\beta_u = \alpha - \theta. \tag{28}$$

The latent variances can be recovered from second moments. Let $\sigma_u^2 := \mathrm{Var}(u)$, $\sigma_v^2 := \mathrm{Var}(v)$, and $\sigma_r^2 := \mathrm{Var}(r)$. Then

$$\sigma_u^2 = \frac{\mathrm{Cov}(W_1, W_2) - \theta\,\mathrm{Var}(W_2)}{\alpha - \theta}, \tag{29}$$

$$\sigma_v^2 = \mathrm{Var}(W_2) - \sigma_u^2, \tag{30}$$

$$\sigma_r^2 = \mathrm{Var}(W_1) - \alpha^2\sigma_u^2 - \theta^2\sigma_v^2. \tag{31}$$

The nonnegativity and plausible sizes of these implied variances are useful model diagnostics.

4.3 How one scalar external instrument can enter

The internal `trigmm` moments do not require an external instrument. If a scalar external instrument Z_{t+1} is available and satisfies the conventional exclusion condition

$$\mathbb{E}\left[Z_{t+1}\{W_{1,t+1} - \theta W_{2,t+1}\}\right] = 0, \tag{32}$$

then it can be added as an additional GMM moment. With one external instrument and the internal moment pair $M_0 = M_1 = 0$, the model is overidentified. This is a particularly useful implementation because it lets the researcher compare three sources of information:

1. the conventional IV estimate from (32);
2. the internal higher-cumulant estimate from (25);
3. the combined GMM estimate using both.

Agreement across these estimates is not proof, but it is a strong robustness check because the identifying assumptions are very different.

4.4 Inference and computation

The point-identified `trigmm` estimator is a nonlinear GMM estimator. Under strong identification and valid moments, standard GMM covariance formulas give standard errors for θ, α, β_u, the variance components, and the recovered X_t coefficients. The paper emphasizes three practical points.

- The objective function can have multiple local minima. Starting values matter, especially with the overidentified $p = 0, 1, 2$ specification.
- The overidentified specification allows a Hansen J test. When the extra moments come from higher cumulants, a rejection indicates a failure of the latent distributional assumptions or the linear model, not merely a bad external instrument.
- Weak or nearly degenerate higher-moment identification shows up as large standard errors and poorly conditioned moment derivatives. The paper suggests checking singular values of the GMM derivative matrix as a diagnostic.

5 Direct option 2 from Lee–Lewbel–Schennach–Zhang: second-moment set identification

The same common-factor model also gives a set-identified result that uses only second moments. This option is useful when the non-Gaussianity needed for Section 4 is not credible or is too weak to estimate precisely.

The underlying Lee–Lewbel–Schennach–Zhang theory contains a sharp identified set based on decomposing the observed distribution into independent Gaussian and non-Gaussian components. That sharp set is conceptually the most complete second-moment object, but it is difficult to estimate and explain in applied work. The Stata paper therefore focuses on a coarser but transparent outer bound based only on the covariance matrix of (W_2, W_1). The formulas below describe that implemented bound.

Assume first that $\beta_u = \alpha - \theta > 0$. Define

$$B_0 := \frac{\mathbb{E}[W_{1,t+1}W_{2,t+1}]}{\mathbb{E}[W_{2,t+1}^2]}, \qquad D_0 := \frac{\mathbb{E}[W_{1,t+1}^2]\mathbb{E}[W_{2,t+1}^2] - \{\mathbb{E}[W_{1,t+1}W_{2,t+1}]\}^2}{\{\mathbb{E}[W_{2,t+1}^2]\}^2}. \tag{33}$$

If the residuals are mean zero, B_0 is simply the population OLS slope from regressing W_1 on W_2, and D_0 is the residual variance from that regression scaled by $\mathrm{Var}(W_2)$.

Because

$$B_0 = \frac{\alpha\mathbb{E}[u^2] + \theta\mathbb{E}[v^2]}{\mathbb{E}[u^2] + \mathbb{E}[v^2]} = \alpha\lambda + \theta(1-\lambda), \qquad \lambda := \frac{\mathbb{E}[u^2]}{\mathbb{E}[u^2] + \mathbb{E}[v^2]} \in [0,1], \tag{34}$$

we obtain the first bound,

$$\theta \le B_0 \le \alpha. \tag{35}$$

The sharper second-moment bound reported in the paper is

$$(B_0 - \theta)(\alpha - B_0) \le D_0. \tag{36}$$

Thus the population identified set is

$$\Theta^{+}_{\mathrm{LSZ,set}} := \{(\alpha, \theta) : \theta \le B_0 \le \alpha,\ (B_0 - \theta)(\alpha - B_0) \le D_0\}. \tag{37}$$

The plus sign indicates $\beta_u > 0$.

If instead $\beta_u < 0$, apply the positive-loading result to $-W_2$. The order reverses and the set becomes

$$\Theta^{-}_{\mathrm{LSZ,set}} := \{(\alpha, \theta) : \alpha \le B_0 \le \theta,\ (\theta - B_0)(B_0 - \alpha) \le D_0\}. \tag{38}$$

5.1 Important consequence: the marginal set for θ is usually not a finite interval

The joint set (37) is informative, but its projection onto θ alone is

$$\{\theta : \exists \alpha \text{ such that } (\alpha, \theta) \in \Theta^{+}_{\mathrm{LSZ,set}}\} = (-\infty, B_0] \tag{39}$$

in population. For $\beta_u < 0$, the projection is $[B_0, \infty)$. Therefore the second-moment set is a joint bound on the causal effect θ and the total common-factor effect α, not a finite scalar interval for θ by itself. A finite interval for θ requires one more substantive restriction, such as a known range for $\beta_u = \alpha - \theta$ or a known range for α.

For example, if $\beta_u > 0$ and $0 \le \beta_u \le \bar{\beta}$, then $\alpha = \theta + \beta_u$ adds

$$0 \le \alpha - \theta \le \bar{\beta}. \tag{40}$$

Projecting the intersection of (37) and (40) gives a finite interval whenever $\bar{\beta}$ is finite. This extra magnitude restriction is not part of the base `trigmmset` second-moment result; it must come from outside the paper.

5.2 Confidence region used by `trigmmset`

In sample, estimate B_0 and D_0 by plug-in estimators $(\widehat{B}, \widehat{D})$ and estimate their covariance matrix $\widehat{V}$ by the delta method. For a significance level δ, construct the ellipse

$$\mathcal{E}_\delta := \left\{ (B, D) : \begin{bmatrix} B - \widehat{B} & D - \widehat{D} \end{bmatrix} \widehat{V}^{-1} \begin{bmatrix} B - \widehat{B} \\ D - \widehat{D} \end{bmatrix} \le \chi^2_{2,1-\delta} \right\}. \tag{41}$$

Then $\mathcal{E}_\delta$ contains the true (B_0, D_0) with approximate probability $1 - \delta$. The command reports a worst-case region that contains the union of all identified sets generated by all $(B, D) \in \mathcal{E}_\delta$. For $\beta_u > 0$, define

$$B^- := \min_{(B,D)\in\mathcal{E}_\delta} B, \qquad B^+ := \max_{(B,D)\in\mathcal{E}_\delta} B, \tag{42}$$

$$D^* := \max_{(B,D)\in\mathcal{E}_\delta} \{D + (B - B^+)(B^- - B)\}. \tag{43}$$

The reported confidence set is the union of four simple regions:

$$\alpha \ge \theta, \quad B^- \le \alpha \le B^+, \quad B^- \le \theta \le B^+, \tag{44}$$

$$B^- \le \alpha \le B^+, \quad \theta \le B^-, \tag{45}$$

$$B^- \le \theta \le B^+, \quad B^+ \le \alpha, \tag{46}$$

$$D^* \ge (\alpha - B^+)(B^- - \theta), \quad B^+ \le \alpha, \quad \theta \le B^-. \tag{47}$$

This construction is conservative but easy to draw and easy to report. It gives a confidence region for the joint object (α, θ). The marginal projection onto θ is again generally one-sided: approximately $(-\infty, B^+]$ for $\beta_u > 0$ and $[B^-, \infty)$ for $\beta_u < 0$.

5.3 Coefficients on controls

The endogenous-equation slope β_2^R is point identified by the projection of Y_2 on X_t. The coefficient vector on X_t in the outcome equation is

$$\beta_1(\theta) = \beta_1^R - \beta_2^R\theta. \tag{48}$$

Because θ is set identified, β_1 is also set identified. The `trigmmset` paper describes a conservative element-by-element way to bound these coefficients using confidence intervals for the components of β_2^R, β_1^R, and B_0. In the scalar notation here, if the sign of a particular component of β_2^R is known, the one-sided bound on θ translates into a one-sided bound on that element of β_1. If the sign of that component is not statistically determined, the corresponding coefficient bound may be uninformative.

6 Options from Lewis's review, translated to the scalar triangular setting

Lewis's Annual Review paper is written for structural vector autoregressions and simultaneous systems with a square impact matrix. It is therefore not identical to the Lewbel triangular model in (1)–(2), and it is also not identical to the Lee–Lewbel–Schennach–Zhang three-shock common-factor model. To use those options in the present scalar problem, impose an alternative square decomposition of the two residuals:

$$U_{t+1} := \begin{bmatrix} W_{2,t+1} \\ W_{1,t+1} \end{bmatrix} = B\eta_{t+1}, \qquad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}. \tag{49}$$

The shocks $\eta_{t+1} = (\eta_{1,t+1}, \eta_{2,t+1})^\top$ are normalized in some way, such as unit variances or a unit diagonal. After estimating B, the scalar coefficient analogous to θ is an impact ratio for a labeled shock. If column j is interpreted as the shock that moves W_2, then

$$\theta_j := \frac{b_{2j}}{b_{1j}}, \qquad b_{1j} \neq 0. \tag{50}$$

This ratio is not automatically the structural θ in (5); it becomes the target only if the researcher is willing to interpret the selected shock as the exogenous variation in W_2 whose causal impact on W_1 is sought. Shock labeling is therefore central.

6.1 Heteroskedasticity option A: two variance regimes

Let the scalar variable Z_{t+1} partition the data into two regimes, say $Z_{t+1} = 0$ and $Z_{t+1} = 1$. Suppose

$$\Sigma_r := \mathrm{Var}(U_{t+1} \mid Z_{t+1} = r) = B\Lambda_r B^\top, \qquad \Lambda_r = \mathrm{diag}(\lambda_{r1}, \lambda_{r2}), \qquad r = 0, 1. \tag{51}$$

The impact matrix B is constant across regimes, but the shock variances change. The two-regime heteroskedasticity condition is

$$\frac{\lambda_{11}}{\lambda_{01}} \neq \frac{\lambda_{12}}{\lambda_{02}}. \tag{52}$$

This says that the two shock variances do not change proportionally across the two regimes. Then

$$\Sigma_1\Sigma_0^{-1} = B\Lambda_1\Lambda_0^{-1}B^{-1}, \tag{53}$$

so the columns of B are eigenvectors of an observable matrix, up to scale and order. Once a column is labeled as the W_2 shock, $\theta_j = b_{2j}/b_{1j}$ is identified.

A particularly transparent special case is that only the variance of the labeled shock j changes. Then

$$\Sigma_1 - \Sigma_0 = (\lambda_{1j} - \lambda_{0j}) b_j b_j^\top, \tag{54}$$

where b_j is the j-th column of B. With $U = (W_2, W_1)^\top$, the impact ratio is

$$\theta_j = \frac{\mathrm{Cov}(W_2, W_1 \mid Z = 1) - \mathrm{Cov}(W_2, W_1 \mid Z = 0)}{\mathrm{Var}(W_2 \mid Z = 1) - \mathrm{Var}(W_2 \mid Z = 0)}. \tag{55}$$

This is the closest variance-regime analogue of a single-instrument estimator. The scalar Z is not an excluded IV; it supplies a high-variance and a low-variance state.

6.2 Heteroskedasticity option B: more regimes, Markov switching, smooth transition, and GARCH

The same covariance-equation logic extends in several directions.

- **More than two regimes.** Additional regimes add more covariance matrices. If B is constant and the diagonal variance changes are rich enough, B is overidentified. Overidentifying restrictions can test whether the fixed impact matrix and diagonal shock-variance assumptions are plausible.

- **Markov-switching variances.** The regimes are latent rather than observed. Identification still comes from distinct variance matrices satisfying a nonproportionality condition, but estimation must also infer the regimes.

- **Smooth transition variances.** The covariance matrix evolves as

  $$\Omega_t = (1 - F(s_t))\Sigma_0 + F(s_t)\Sigma_1, \tag{56}$$

  where s_t is a transition variable. Conditional on the transition parameters, the endpoints Σ_0 and Σ_1 identify B by the same logic as Section 6.1. Identification is weak if the transition is nearly flat or if s_t does not track actual volatility changes.

- **GARCH or stochastic volatility.** Parametric volatility models use the full time path of conditional variances. The maintained idea remains fixed B plus diagonal, time-varying structural variances. Parametric assumptions can improve efficiency but introduce functional-form risk.

In all of these variants, one can recover an impact ratio such as (50) only after normalizing and labeling the relevant column of B.

6.3 Heteroskedasticity option C: unconditional moments of squared residuals

Lewis also reviews distribution-free identification based on persistence in volatility. In the present two-variable notation, the useful objects are autocovariances of squared and cross-product residuals, such as

$$\mathrm{Cov}\left(\mathrm{vec}(U_{t+1}U_{t+1}^\top), \mathrm{vec}(U_{t+1-s}U_{t+1-s}^\top)^\top\right), \qquad s > 0. \tag{57}$$

If structural shocks themselves are serially uncorrelated but their variances are persistent, these autocovariances reveal how the structural variances move over time. Identification requires the volatility processes of the shocks not to have proportional persistence patterns. This option is attractive because it does not require choosing exact regimes, but it is demanding in finite samples because it uses fourth-order quantities and their autocovariances.

6.4 Non-Gaussianity option A: independent component analysis

Instead of changing variances over time, assume that the components of η_{t+1} in (49) are mutually independent and that at most one of them is Gaussian. Then, by the Darmois–Skitovich/Comon logic reviewed by Lewis, the decomposition $U = B\eta$ is unique up to scale, sign, and column order. In the two-shock case this means that at least one shock must be non-Gaussian.

Independent component analysis (ICA) estimates the rotation of the whitened residual vector that makes the candidate shocks as independent or as non-Gaussian as possible. After estimating B, the scalar causal parameter is again an impact ratio b_{2j}/b_{1j} for the economically labeled column. The single scalar Z can help label the column, for example by choosing the shock most correlated with Z or the shock whose response signs match prior theory, but the statistical identification itself comes from independence and non-Gaussianity, not from the exclusion restriction $\mathbb{E}[Z\varepsilon_1] = 0$.

6.5 Non-Gaussianity option B: likelihood and pseudo-likelihood

A parametric likelihood version specifies densities for the independent shocks, for example Student-t densities. If the density is correctly specified, maximum likelihood can be efficient. Pseudo-maximum likelihood can remain consistent under certain misspecifications, provided the pseudo-model is itself identified. The cost is that there are many possible ways to be non-Gaussian, so a particular density choice is an additional modeling decision.

In the scalar triangular report, this option again delivers θ only through an estimated square mixing matrix and a labeled impact ratio. It is not the same as the Lee–Lewbel–Schennach–Zhang cumulant equations, which are tailored to the three-component common-factor model.

6.6 Non-Gaussianity option C: moment-based GMM with coskewness or cokurtosis

Moment-based non-Gaussian identification selects particular independence-implied moments. Let $\eta = CU$ denote candidate shocks after applying an unmixing matrix $C = B^{-1}$ and normalizing variances to one. Typical restrictions include

$$\mathbb{E}[\eta_1\eta_2] = 0, \tag{58}$$

$$\mathbb{E}[\eta_1^2\eta_2] = 0, \qquad \mathbb{E}[\eta_1\eta_2^2] = 0, \tag{59}$$

$$\mathbb{E}[\eta_1^2\eta_2^2] - 1 = 0, \qquad \mathbb{E}[\eta_1^3\eta_2] = 0, \qquad \mathbb{E}[\eta_1\eta_2^3] = 0. \tag{60}$$

The restrictions in (59) use skewness-type information; those in (60) use kurtosis-type information. Identification requires at least one shock to have relevant nonzero skewness or nonzero excess kurtosis. A GMM estimator chooses C so these cross-moments are close to zero, then obtains $B = C^{-1}$ and the impact ratio (50).

This option is transparent because the identifying moments are explicit, but it can be noisy: estimating fourth moments and the covariance matrix of moment conditions may require moments up to order eight.

6.7 Non-independent components and coheteroskedasticity

The Annual Review also discusses newer work relaxing full independence. The core idea is to replace full independence by restrictions on higher-order cumulant tensors, such as diagonal cumulant tensors or reflectional-invariance patterns. These methods are promising because they can allow some forms of dependence, including coheteroskedasticity. In the scalar triangular problem they are best viewed as advanced alternatives to the simple square ICA model, not as direct Lewbel or `trigmm` estimators.

7 Using one scalar instrument with each option

The same observed scalar Z can play different roles. Confusing these roles is the easiest way to misuse these methods.

Role 1. External IV. If $\mathbb{E}[Z\varepsilon_1] = 0$ and $\mathbb{E}[ZW_2] \neq 0$, then Z directly identifies θ by conventional IV:

$$\theta_{\mathrm{IV}} = \frac{\mathbb{E}[ZW_1]}{\mathbb{E}[ZW_2]}. \tag{61}$$

This moment can be combined with `trigmm`'s internal moments.

Role 2. Lewbel heteroskedasticity shifter. If Z satisfies (8), then $(Z - \mathbb{E}Z)W_2$ is the constructed instrument. Here the relevant covariance is $\mathrm{Cov}(Z, W_2^2)$, not $\mathbb{E}[ZW_2]$.

Role 3. Variance-regime indicator or volatility state. If Z defines high- and low-volatility regimes, then it supplies two covariance matrices. The assumption is not exclusion but fixed B and nonproportional changes in shock variances.

Role 4. Shock-labeling variable. In ICA or non-Gaussian square models, Z may help decide which column of B is economically meaningful. It does not by itself identify the rotation unless an additional IV or moment condition is imposed.

Role 5. No role in internal LSZ identification. The Lee–Lewbel–Schennach–Zhang point and set identification results do not require Z. They can be used when no external instrument exists. A valid Z can be added, but it is optional.

8 Inference and weak-identification issues

All the methods in this report use information beyond the first two unconditional moments of a homoskedastic linear regression. That information can be weak.

Lewbel exact and bounded-invalidity inference. If $\mathrm{Cov}(Z, W_2^2)$ is near zero, the constructed instrument is weak. Standard IV weak-instrument diagnostics are informative because the estimator is implemented as two-stage least squares with the constructed instrument. In the bounded invalidity interval, the same weak relevance makes the interval wide and can make sample roots unstable.

Lee–Lewbel–Schennach–Zhang point inference. If the relevant higher cumulants of u and v are small or nearly degenerate, the GMM objective is flat in some direction. Then standard errors can be large, local minima become a practical problem, and Wald intervals can be misleading. The overidentified $p = 0, 1, 2$ specification, the Hansen J test, and singular values of the moment derivative matrix are useful diagnostics, but none proves identification strength in finite samples.

Lee–Lewbel–Schennach–Zhang set inference. The `trigmmset` confidence region is explicitly set-valued. It first builds an ellipse for (B_0, D_0) and then reports a conservative worst-case joint region for (α, θ). This is more honest than reporting a spurious finite scalar confidence interval for θ when the population identified set itself projects to a half-line.

Variance-regime inference. If variances change little, or if both shock variances change almost proportionally, the eigenvectors in (53) are weakly identified. Robust GMM confidence sets or weak-identification-robust procedures are preferable to simple Wald intervals.

Non-Gaussian inference. If the shocks are close to Gaussian, independence-based rotations are weakly identified. Testing for non-Gaussianity and independence is useful but difficult because the shocks are estimated objects. Robust score/projection methods have been developed in the literature, but they are more complex than ordinary standard errors.

9 Software and implementation notes

- **Stata.** Lee et al. provide `trigmm` for the point-identified non-Gaussian common-factor method and `trigmmset` for the second-moment set. The commands are wrappers around Stata's `gmm` infrastructure and use Stata post-estimation tools. Lewbel's heteroskedasticity-generated instruments are implemented in `ivreg2h`.
- **R or Python.** The scalar formulas in Sections 3 through 5 are simple enough to implement directly. For the LSZ point estimator, one needs routines for sample cumulants and nonlinear GMM. For set estimation, one needs only plug-in estimates of B_0 and D_0, their delta-method covariance matrix, and a routine to draw or optimize over the reported joint region. Generic GMM, nonlinear optimization, and bootstrap tools are sufficient, but a direct off-the-shelf implementation of `trigmm` for R is not part of the attached papers.
- **ICA and SVAR methods.** Generic ICA, VAR, GARCH, and structural-VAR software can estimate some of the broader Lewis-type models. The researcher must still map the estimated impact matrix into the scalar ratio (50) and justify the shock label. These packages do not automatically estimate the Lewbel triangular coefficient θ.

10 A practical decision tree

1. **Start with the maintained triangular model.** Residualize both $Y_{1,t+1}$ and $Y_{2,t+1}$ on the same X_t and work with $W_{1,t+1}, W_{2,t+1}$.
2. **If a conventional scalar IV is credible, use it and report it.** Estimate (61) or its controlled version. Then consider adding Lewbel or `trigmm` moments as robustness checks or efficiency-improving overidentifying restrictions.
3. **If no conventional IV is credible but a heteroskedasticity shifter is credible, use Lewbel.** Check whether $\mathrm{Cov}(Z, W_2^2)$ is meaningfully nonzero. If exact validity is doubtful, report the bounded-invalidity interval over a grid of τ values.
4. **If the common-factor story is credible and the errors are plausibly non-Gaussian, use `trigmm`.** Specify the sign of β_u, estimate with $p = 0, 1$ and with $p = 0, 1, 2$, vary starting values, and report diagnostics.
5. **If the common-factor story is credible but non-Gaussianity is not, use `trigmmset`.** Report the joint region for (α, θ) and state clearly whether the implied marginal set for θ is one-sided. Do not report a finite interval for θ unless an additional magnitude restriction is imposed.

6. **If the data are time series with credible volatility regimes or persistent volatility, consider the Lewis heteroskedasticity options.** Make clear that this replaces the common-factor model by a square impact-matrix model and that θ is an impact ratio for a labeled shock.

7. **If non-Gaussian independent shocks are credible, consider ICA, likelihood, or higher-moment GMM.** Again, report how the shock was labeled and how the impact ratio was normalized.

11 Conclusion

In the scalar triangular setting, the methods differ mainly in what replaces the missing excluded instrument. Lewbel uses a scalar heteroskedasticity shifter to construct an instrument. Lee–Lewbel–Schennach–Zhang use a common-factor error structure: higher cumulants yield point identification, while second moments yield a conservative joint set. Lewis's review offers a menu of broader higher-moment strategies–variance regimes, volatility processes, ICA, likelihood, and moment-based non-Gaussian GMM–but those strategies require a square shock decomposition and a shock-labeling step before they deliver a scalar effect comparable to θ.

The most transparent reporting strategy is to state, for each estimate or set, which object identifies θ: an exclusion moment, a heteroskedasticity covariance, a non-Gaussian cumulant system, a second-moment common-factor inequality, a variance-regime covariance change, or an independent-component rotation. Once that distinction is clear, all other coefficients follow from the simple recovery formula $\beta_1(w) = \beta_1^R - \beta_2^R w$.

12 Inference and R bootstrap implementation guide

The preceding sections describe what identifies the coefficient of the scalar endogenous regressor. This section describes how to attach standard errors, confidence intervals, or confidence sets to the resulting estimate or identified set. The emphasis is deliberately practical: what can be done directly in R, what requires a short custom wrapper, and what is currently easier to do in Stata or by translating published Stata/MATLAB code.

There are three distinct inferential targets. They should not be mixed.

1. *Point-identified inference*: report a standard error or confidence interval for a singleton estimate $\hat{\theta}$.
2. *Endpoint inference*: report uncertainty about the lower and upper endpoints (L_0, U_0) of a scalar identified interval $\Theta_0 = [L_0, U_0]$.
3. *Partial-identification confidence sets*: report a set that covers either the entire identified set Θ_0 or the true unknown scalar value $\theta_0 \in \Theta_0$. These are different coverage goals.

In the scalar triangular setting, the full model often reduces to a one-dimensional coefficient set for θ. Once a candidate w is retained, the other coefficients are recovered from

$$\beta_1(w) = \beta_1^R - \beta_2^R w. \tag{62}$$

Thus the most transparent workflow is: first form an estimate or confidence set for θ, and then map each endpoint or retained grid value through the coefficient recovery formula.

12.1 What was found in R

Table 2 summarizes the R packages and code bases that are most relevant to the current problem. The important conclusion is that R has usable point-estimation and bootstrap building blocks, but I did not find a standard R package that directly implements the Lewbel (2012) Theorem 3 set-identified interval or the Lee–Lewbel–Schennach–Zhang `trigmmset` confidence region.

Table 2: R packages and code bases relevant to bootstrap inference in the scalar triangular model.

Package or code base	Role in this problem	What it provides	Limitation
`REndo::hetErrorsIV()`	Closest direct R implementation of the Lewbel heteroskedastic-error triangular estimator.	Estimates linear models with endogenous regressors using heteroskedastic covariance restrictions; reports robust and cluster-adjusted standard errors.	Point-identified estimator only; no direct bootstrap argument for the set-identified bounds.
`ivlewbel::lewbel()`	Older direct Lewbel-style R implementation.	Implements heteroskedasticity-based GMM/IV estimation for triangular models.	Older and less maintained; point-estimation focus.

Package or code base	Role in this problem	What it provides	Limitation
`ivreg`	General IV/2SLS infrastructure.	After manually constructing Lewbel instruments, runs 2SLS and exposes an object compatible with many covariance and bootstrap tools.	Does not construct Lewbel instruments or set bounds automatically.
`gmm`	User-supplied moment conditions.	Natural framework for hand-coded nonlinear `trigmm`-style moments, Lewbel moments, and custom GMM covariance estimation.	Bootstrap and set-inversion must be wrapped by the user.
`boot`	General full-procedure bootstrap.	Resample rows, clusters, or blocks; return any statistic, including a point estimate, a lower endpoint, an upper endpoint, or a whole vector of moments.	Validity depends on the statistic; ordinary n-out-of-n bootstrap can fail near nonsmooth set boundaries.
`sandwich::vcovBS()`	Generic bootstrap covariance for model objects.	Case or cluster bootstrap covariance matrices for supported objects through `update()` and `coef()`.	Less convenient for custom endpoint sets that are not model objects.
`fwildclusterboot`	Wild cluster bootstrap for linear and IV models.	Supports `ivreg` objects, so it can be used after manual construction of Lewbel instruments.	Point-identified IV inference; not a partial-identification set package.
`clusterSEs::cluster.wild.ivreg()`	Alternative wild cluster bootstrap for IV.	Wild-cluster-bootstrap p-values and intervals for `ivreg` models.	Point-identified IV inference only.
`ivmodel`	Weak-IV robust inference with one endogenous regressor.	Anderson–Rubin, conditional likelihood ratio, and related intervals for linear IV.	Designed for conventional linear IV, not for `trigmmset` or Lewbel Theorem 3 set endpoints.
`moonboot`	m-out-of-n bootstrap building block.	Useful when ordinary bootstrap is suspect for nonsmooth or irregular statistics.	General method; no triangular-system estimator.

Package or code base	Role in this problem	What it provides	Limitation
`ivmte::boundCI()`	Useful R example of bootstrap inference for partial-identification bounds.	Takes original bounds and bootstrapped bounds and constructs confidence intervals for bounds.	For marginal treatment effect bounds, not for the scalar triangular model; useful mainly as a template.

The practical R stack is therefore:

`REndo` or `ivreg` for point estimation, `boot` for full-procedure resampling, `gmm` for custom moments.

For clustered data, replace row resampling by cluster resampling or use a wild cluster bootstrap after the generated instrument has been constructed.

12.2 Point-identified Lewbel estimator: full-procedure bootstrap

For Lewbel's exact point-identified triangular estimator, the bootstrap must repeat the entire generated-instrument construction. It is not enough to hold $\hat{\varepsilon}_2$ fixed across bootstrap draws.

For each bootstrap draw $b = 1, \ldots, B$:

1. resample observations, clusters, or time blocks according to the sampling design;
2. regress Y_2 on X in the bootstrap sample and compute $\hat{\varepsilon}^{*(b)}_{2,i}$;
3. construct $H^{*(b)}_i = (Z^{*(b)}_i - \bar{Z}^{*(b)})\hat{\varepsilon}^{*(b)}_{2,i}$;
4. estimate the structural equation by IV using $X^{*(b)}$ and $H^{*(b)}$ as instruments;
5. store $\hat{\theta}^{*(b)}$ and any recovered coefficient $\hat{\beta}^{*(b)}_1 = \hat{\beta}^{R,*(b)}_1 - \hat{\beta}^{R,*(b)}_2 \hat{\theta}^{*(b)}$.

A percentile interval is the interval between the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of the bootstrap draws. A studentized interval stores $(\hat{\theta}^{*(b)} - \hat{\theta})/\widehat{\mathrm{se}}^{*(b)}$ and inverts the resulting quantiles. A basic interval reflects bootstrap endpoint deviations around the original estimate.

A minimal R skeleton is:

```
# install.packages(c("ivreg", "boot"))
library(ivreg)
library(boot)

lewbel_point <- function(dat, y1, y2, X, Z) {
  f2 <- reformulate(X, response = y2)
  rf <- lm(f2, data = dat)
  dat$eps2_hat <- resid(rf)
  dat$H_lewbel <- (dat[[Z]] - mean(dat[[Z]], na.rm = TRUE)) * dat$eps2_hat

  rhs <- paste(c(X, y2), collapse = " + ")
  inst <- paste(c(X, "H_lewbel"), collapse = " + ")
```

```
  fiv <- as.formula(paste(y1, "~", rhs, "|", inst))
  fit <- ivreg(fiv, data = dat)
  unname(coef(fit)[y2])
}

boot_lewbel_point <- function(dat, y1, y2, X, Z, R = 1999, seed = 123) {
  set.seed(seed)
  stat <- function(d, idx) lewbel_point(d[idx, , drop = FALSE], y1, y2, X, Z)
  out <- boot(dat, statistic = stat, R = R)
  list(theta_hat = out$t0,
       percentile = boot.ci(out, type = "perc"),
       basic      = boot.ci(out, type = "basic"),
       boot_obj   = out)
}
```

For clustered data, replace the observation index by a cluster index. For a wild cluster bootstrap, a convenient workflow is to construct the Lewbel instrument in the original data, estimate the IV model with `ivreg`, and then apply `fwildclusterboot::boottest()` to the coefficient of Y_2. This is a point-identified IV bootstrap; it does not bootstrap the Lewbel set-identified interval.

12.3 Lewbel Theorem 3: bootstrapping a scalar set-identified interval

Lewbel's Theorem 3 relaxes exact validity by allowing $\mathrm{Cov}(Z, \varepsilon_1\varepsilon_2)$ to be nonzero but bounded. In the scalar case, suppose that for a user-chosen $\tau \in [0, 1)$,

$$|\,\mathrm{Corr}(Z, \varepsilon_1\varepsilon_2)| \le \tau |\,\mathrm{Corr}(Z, \varepsilon_2^2)|. \tag{63}$$

Let W_1 and W_2 be the residuals from regressions of Y_1 and Y_2 on X. Define the sample analogues of

$$\begin{aligned} C_{12,Z} &= \mathrm{Cov}(W_1W_2, Z), & C_{22,Z} &= \mathrm{Cov}(W_2^2, Z),\\ V_{12} &= \mathrm{Var}(W_1W_2), & V_{22} &= \mathrm{Var}(W_2^2),\\ C_{12,22} &= \mathrm{Cov}(W_1W_2, W_2^2). \end{aligned}$$

The estimated endpoints of the Lewbel interval are the two real roots of

$$A\theta^2 + B\theta + C = 0, \tag{64}$$

where

$$A = 1 - \tau^2, \tag{65}$$

$$B = 2\left(\frac{C_{12,22}}{V_{22}}\tau^2 - \frac{C_{12,Z}}{C_{22,Z}}\right), \tag{66}$$

$$C = \left(\frac{C_{12,Z}}{C_{22,Z}}\right)^2 - \frac{V_{12}}{V_{22}}\tau^2. \tag{67}$$

When $\tau = 0$, the two roots collapse to the exact Lewbel point estimate $C_{12,Z}/C_{22,Z}$.

The full-procedure bootstrap for this set is:

1. compute the original interval $[\hat{L}, \hat{U}]$ from the original data;

2. resample observations, clusters, or blocks;
3. in each draw, recompute W_1^*, W_2^*, all covariance terms, and the two roots $[\hat{L}^*, \hat{U}^*]$;
4. build endpoint intervals or a partial-identification confidence set from the empirical distribution of $(\hat{L}^*, \hat{U}^*)$.

A minimal R implementation is:

```
pop_cov <- function(a, b) {
  mean((a - mean(a, na.rm = TRUE)) *
       (b - mean(b, na.rm = TRUE)), na.rm = TRUE)
}

rhs_formula <- function(y, X) {
  if (length(X) == 0) as.formula(paste(y, "~ 1")) else reformulate(X, response = y)
}

lewbel_set_bounds <- function(dat, y1, y2, X, Z, tau, eps = 1e-10) {
  stopifnot(tau >= 0, tau < 1)

  W1 <- resid(lm(rhs_formula(y1, X), data = dat))
  W2 <- resid(lm(rhs_formula(y2, X), data = dat))
  z  <- dat[[Z]]

  A12 <- W1 * W2
  A22 <- W2^2

  c12z  <- pop_cov(A12, z)
  c22z  <- pop_cov(A22, z)
  v12   <- pop_cov(A12, A12)
  v22   <- pop_cov(A22, A22)
  c1222 <- pop_cov(A12, A22)

  if (!is.finite(c22z) || abs(c22z) < eps ||
      !is.finite(v22) || v22 < eps) {
    return(c(L = NA_real_, U = NA_real_))
  }

  A <- 1 - tau^2
  B <- 2 * ((c1222 / v22) * tau^2 - (c12z / c22z))
  C <- (c12z / c22z)^2 - (v12 / v22) * tau^2

  disc <- B^2 - 4 * A * C
  if (!is.finite(disc) || disc < 0) return(c(L = NA_real_, U = NA_real_))

  roots <- sort((-B + c(-1, 1) * sqrt(disc)) / (2 * A))
  c(L = roots[1], U = roots[2])
}
```

```r
bootstrap_lewbel_set <- function(dat, y1, y2, X, Z, tau,
                                 R = 1999, alpha = 0.05,
                                 m = nrow(dat), replace = TRUE,
                                 seed = 123) {
  set.seed(seed)
  n <- nrow(dat)

  theta_hat <- lewbel_set_bounds(dat, y1, y2, X, Z, tau)

  reps <- replicate(R, {
    ii <- sample.int(n, size = m, replace = replace)
    lewbel_set_bounds(dat[ii, , drop = FALSE], y1, y2, X, Z, tau)
  })

  reps <- t(reps)
  reps <- reps[complete.cases(reps), , drop = FALSE]

  list(
    point_bounds = theta_hat,
    percentile_outer = c(
      L = unname(quantile(reps[, "L"], alpha / 2, na.rm = TRUE)),
      U = unname(quantile(reps[, "U"], 1 - alpha / 2, na.rm = TRUE))
    ),
    basic_outer = c(
      L = 2 * theta_hat["L"] -
          unname(quantile(reps[, "L"], 1 - alpha / 2, na.rm = TRUE)),
      U = 2 * theta_hat["U"] -
          unname(quantile(reps[, "U"], alpha / 2, na.rm = TRUE))
    ),
    boot_reps = reps
  )
}
```

The interval named `percentile_outer` is easy to explain: it expands the estimated lower endpoint downward and the estimated upper endpoint upward using the empirical endpoint distributions. This is a confidence interval for the estimated set's endpoints, not automatically an optimal Imbens–Manski interval for the unknown scalar $\theta_0 \in [L_0, U_0]$. For a paper, report which coverage target is being used.

12.4 Imbens–Manski/Stoye endpoint inference

When the scalar identified set is $\Theta_0 = [L_0, U_0]$ and the endpoints are estimated by $(\hat{L}, \hat{U})$, the most common partial-identification interval has the form

$$\left[\hat{L} - c_n \widehat{\mathrm{se}}(\hat{L}), \quad \hat{U} + c_n \widehat{\mathrm{se}}(\hat{U})\right]. \tag{68}$$

For set coverage, a conservative choice is the Bonferroni value $c_n = z_{1-\alpha/2}$, or the projection of a joint bootstrap confidence region for (L_0, U_0). For coverage of the true scalar point $\theta_0 \in [L_0, U_0]$, the Imbens–Manski critical value is generally smaller and depends on the estimated width of the identified set relative to endpoint uncertainty. Stoye's refinement is important near point identification because naive endpoint intervals can have nonuniform coverage when $U_0 - L_0$ is close to zero.

A practical algorithm using the bootstrap draws from Section 12.3 is:

1. estimate $[\hat{L}, \hat{U}]$ and bootstrap draws $[\hat{L}^{*(b)}, \hat{U}^{*(b)}]$;
2. estimate the covariance matrix of $(\hat{L}, \hat{U})$ from the bootstrap draws;
3. for a conservative interval, project the bootstrap joint region or use the Bonferroni outer interval;
4. for an Imbens–Manski/Stoye interval, solve the published critical value equation using the endpoint standard errors and the estimated width $\hat{U} - \hat{L}$.

This method is most appropriate for Lewbel Theorem 3, because that theorem gives a scalar interval directly. It is less natural for `trigmmset` unless the joint (α, γ) set has first been projected to a finite scalar interval for γ.

12.5 A `trigmmset`-style second-moment bootstrap in R

Lee–Lewbel–Schennach–Zhang's `trigmmset` command uses two observable second-moment objects. In the no-covariate residualized notation of this report,

$$B_0 = \frac{\mathbb{E}(W_1 W_2)}{\mathbb{E}(W_2^2)}, \qquad D_0 = \frac{\mathbb{E}(W_1^2)\mathbb{E}(W_2^2) - \mathbb{E}(W_1 W_2)^2}{\mathbb{E}(W_2^2)^2}. \tag{69}$$

For positive common-factor loading, the structural parameters satisfy

$$\gamma \le B_0 \le \alpha, \qquad (B_0 - \gamma)(\alpha - B_0) \le D_0. \tag{70}$$

The Stata command estimates (B_0, D_0), uses a delta-method covariance matrix, forms an ellipse in (B, D) space, and reports a conservative worst-case region for (α, γ) over that ellipse.

In R, a simple bootstrap building block is to resample $(\hat{B}, \hat{D})$:

```
trigmmset_BD <- function(dat, w, y) {
  ww <- dat[[w]]
  yy <- dat[[y]]

  Ey2 <- mean(yy^2, na.rm = TRUE)
  Ewy <- mean(ww * yy, na.rm = TRUE)
  Ew2 <- mean(ww^2, na.rm = TRUE)

  B <- Ewy / Ey2
  D <- (Ew2 * Ey2 - Ewy^2) / Ey2^2

  c(B = B, D = D)
}

bootstrap_trigmmset_BD <- function(dat, w, y, R = 1999,
                                   alpha = 0.05, seed = 123) {
  set.seed(seed)
  n <- nrow(dat)
```

```
  bd_hat <- trigmmset_BD(dat, w, y)

  reps <- replicate(R, {
    ii <- sample.int(n, n, replace = TRUE)
    trigmmset_BD(dat[ii, , drop = FALSE], w, y)
  })

  reps <- t(reps)

  list(
    BD_hat = bd_hat,
    BD_percentile_CI = apply(
      reps, 2, quantile,
      probs = c(alpha / 2, 1 - alpha / 2),
      na.rm = TRUE
    ),
    BD_cov = cov(reps, use = "complete.obs"),
    BD_reps = reps
  )
}
```

This code only bootstraps (B_0, D_0). To reproduce the logic of `trigmmset`, one must turn the joint uncertainty in (B, D) into a joint confidence region and then map that region through the inequalities in (70). A bootstrap analogue of the Stata ellipse is:

1. compute $\hat{\mu} = (\hat{B}, \hat{D})^\top$ and bootstrap draws $\hat{\mu}^{*(b)} = (\hat{B}^{*(b)}, \hat{D}^{*(b)})^\top$;
2. compute a covariance matrix $\hat{V}_{BD}$ from the bootstrap draws;
3. form the ellipsoid
$$\mathcal{E}_\alpha^{boot} = \left\{ \mu : (\mu - \hat{\mu})^\top \hat{V}_{BD}^{-1} (\mu - \hat{\mu}) \leq c_{1-\alpha}^{boot} \right\}, \tag{71}$$
where $c_{1-\alpha}^{boot}$ is the empirical $1 - \alpha$ quantile of $(\hat{\mu}^{*(b)} - \hat{\mu})^T \hat{V}_{BD}^{-1} (\hat{\mu}^{*(b)} - \hat{\mu})$;
4. report the union of all structural sets implied by $(B, D) \in \mathcal{E}_\alpha^{boot}$.

If the object of interest is only γ, report the projection of that union onto the γ axis. Under only (70), that projection can be one-sided or unbounded. A finite interval for γ requires an extra restriction, for example a bounded range for the common-factor loading $\beta = \alpha - \gamma$.

12.6 Moment-inequality and test-inversion bootstrap

Many set-identified confidence procedures can be expressed as test inversion. For each candidate value w of θ, define a null hypothesis $H_0 : \theta = w$. Compute a statistic $T_n(w)$ that measures whether the sample moments or inequalities are violated at w. Retain w if the test does not reject. The confidence set is
$$\mathcal{C}_{1-\alpha} = \{w : T_n(w) \leq c_{1-\alpha}(w)\}. \tag{72}$$

For a scalar moment-inequality representation, a common statistic is
$$T_n(w) = \max_{j \leq J} \frac{\sqrt{n}\, \bar{m}_j(w)}{\hat{\sigma}_j(w)}, \tag{73}$$

where the null requires $\mathbb{E}[m_j(w)] \le 0$ for all j. The critical value can be computed by generalized moment selection, subsampling, or an m-out-of-n bootstrap. The R implementation is necessarily custom unless one can adapt a specialized package from a different application.

The generic R loop is:

```
theta_grid <- seq(theta_min, theta_max, length.out = 1000)
keep <- logical(length(theta_grid))

for (k in seq_along(theta_grid)) {
  w <- theta_grid[k]

  # 1. compute sample moment inequalities mbar(w)
  # 2. compute a studentized test statistic Tn_w
  # 3. get a critical value by bootstrap, m-out-of-n bootstrap, or subsampling
  # 4. retain w if not rejected
  keep[k] <- (Tn_w <= c_alpha_w)
}

ci_theta <- range(theta_grid[keep])
```

This approach is more work than bootstrapping endpoints, but it is conceptually clean when the scalar coefficient is a projection of a higher-dimensional identified set, as in `trigmmset`.

12.7 Subsampling and m-out-of-n bootstrap

Set endpoints are often nonsmooth functions of sample moments: a discriminant can cross zero, a constraint can switch from slack to binding, or a projection can move from bounded to unbounded. In such cases, the ordinary n-out-of-n bootstrap may be unreliable. Two robust alternatives are:

1. *subsampling*: draw subsets of size b without replacement, with $b \to \infty$ and $b/n \to 0$;
2. *m-out-of-n bootstrap*: draw $m < n$ observations with replacement, again with $m \to \infty$ and $m/n \to 0$.

For implementation, replace the line

```
ii <- sample.int(n, n, replace = TRUE)
```

by either

```
ii <- sample.int(n, b, replace = FALSE)  # subsampling
```

or

```
ii <- sample.int(n, m, replace = TRUE)   # m-out-of-n bootstrap
```

The statistic must then be scaled appropriately. For an endpoint statistic, compare $\sqrt{b}(\hat{L}_b - \hat{L})$ or $\sqrt{m}(\hat{L}^*_m - \hat{L})$ to $\sqrt{n}(\hat{L} - L_0)$. In applied work, report sensitivity across a few values of b or m rather than relying on one arbitrary tuning value.

12.8 Recovering confidence intervals for the other coefficients

Because $\beta_1(w) = \beta_1^R - \beta_2^R w$, a set or interval for θ maps into a set for each coefficient. If $\theta \in [L, U]$, the kth coefficient is in

$$\left\{\beta_{1,k}^R - \beta_{2,k}^R w : w \in [L, U]\right\}. \tag{74}$$

If $\beta_{2,k}^R > 0$, this interval is

$$[\beta_{1,k}^R - \beta_{2,k}^R U, \beta_{1,k}^R - \beta_{2,k}^R L]. \tag{75}$$

If $\beta_{2,k}^R < 0$, the endpoints are reversed. In a bootstrap, the safest procedure is to recompute the residualization and recovery map in each draw:

$$\beta_{1,k}^{*(b)}(w) = \beta_{1,k}^{R,*(b)} - \beta_{2,k}^{R,*(b)} w. \tag{76}$$

This automatically accounts for the sampling uncertainty in the reduced-form coefficients and in the identified interval endpoints.

12.9 Recommended reporting template

For the scalar triangular model, a clear empirical table should report:

1. the OLS slope of W_1 on W_2;
2. the exact Lewbel point estimate, if the exact covariance condition is maintained;
3. first-stage relevance for the generated instrument, e.g. $\widehat{\mathrm{Cov}}(Z, W_2^2)$ and a heteroskedasticity test;
4. the Lewbel bounded-invalidity interval for several values of τ;
5. bootstrap endpoint intervals for each reported set;
6. if using the common-factor method, the `trigmm` point estimate or the `trigmmset`-style joint region and its projection for θ;
7. a statement of the coverage target: endpoint coverage, identified-set coverage, or true-parameter coverage.

A short but rigorous sentence is: "The bootstrap resamples observations and re-estimates the residualizations, generated instruments, and set endpoints in each replication; the reported interval is an outer endpoint interval for the identified set."

13 Practical conclusions for software choice

The most direct software choice depends on the identification target. For Lewbel point identification in R, use `REndo::`
`hetErrorsIV()` or manually construct the generated instrument and use `ivreg`. For bootstrap inference, manually wrap the whole construction in `boot`. For clustered point-identified IV inference, use `fwild`
`clusterboot` or `clusterSEs` after constructing the generated instrument.

For Lewbel's Theorem 3 set-identified interval, no standard R package was found that automates the method. The necessary computation is short: compute residuals, solve the quadratic for the two endpoints, and bootstrap the full procedure. The code in Section 12.3 is the most direct R implementation.

For the Lee–Lewbel–Schennach–Zhang common-factor model, Stata remains the direct implementation: `trigmm` for point identification and `trigmmset` for the second-moment set bound. In R, `gmm` can be used to code the nonlinear moments for point identification, while a `trigmmset`-style set analysis requires custom code for the (B, D) confidence region and the projection step.

For publication-quality partial-identification inference, the most credible approach is to report both a transparent endpoint-bootstrap interval and a sensitivity check using subsampling or m-out-of-n bootstrap when endpoints are unstable. If the scalar parameter is a projection of a multidimensional moment-inequality set, use test inversion or calibrated projection rather than a naive percentile interval.

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